Rick Akkerman

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Computer vision engineer and researcher with 3+ years of experience in 2D/3D computer vision. I have experience building 3D digital-human systems, most recently as a Research Associate in Michael J. Black's research group at the Max Planck Institute for Intelligent Systems, Tübingen. My work centers on parametric body and face models (SMPL-X, MANO, FLAME), diffusion models, and vision-language models (VLM). This resulted in two CVPR 2025 papers, one as a first author and one as a contributor. I work in PyTorch across the full pipeline, from data capture and generation through rendering, to model fine-tuning and evaluation. Throughout internships, I have also gained experience with computer vision-based quality inspection and thermography.

Experience

Max Planck Institute for Intelligent Systems, Perceiving Systems

Research Associate

02/2024 – Present *Tübingen, Germany*

- First-authored InterDyn (CVPR 2025): fine-tuned video diffusion models to generate controllable object dynamics from a driving signal, showing large video models act as implicit neural physics engines
- Contributed to OFER (CVPR 2025): diffusion-based 3D face (FLAME) reconstruction from a single, heavily occluded image; captures the problem's multi-hypothesis nature by generating a range of plausible expressions instead of one
- Designed and built an end-to-end training pipeline for vision-language-action (VLA) models on the EgoExo4D dataset. Optimized data storage and data loaders, and ran distributed training and evaluation on H100 GPUs (unpublished)
- Engineered an agentic navigation system pairing a frozen VLM with a motion-diffusion prior to drive SMPL-X avatars through scanned 3D scenes (ScanNet) (internal)
- Optimized a 3D face-fitting pipeline for FLAME, cutting per-fit runtime while improving reconstruction (internal)

Loyens & Loeff

Working Student, Technology & Innovation

06/2023 – 08/2023 *Amsterdam, The Netherlands*

Supported the adoption of AI-powered legal-tech tools (including automated drafting) at a leading law firm. I also researched the legal-AI market to advise on solution selection for specific business cases.

Innovation Centre for Artificial Intelligence

Communication Manager (self-employed)

08/2022 – 03/2023 *Amsterdam, The Netherlands*

Grew the public profile of UvA's AI innovation center. I led interview campaigns with AI researchers, helped organise national AI events, and produced and distributed all web, LinkedIn, and social content.

Perron038

Machine Learning Internship

02/2021 – 05/2021 *Zwolle, The Netherlands*

Prototyped a non-destructive method to detect leakage invisible from the outside of sealed detergent-pod containers using active thermography. Co-designed a cobot-driven capture-and-heating rig for repeatable imaging, captured a thermal image dataset, and demonstrated reliable leak detection.

Tembo Group

Design Science Internship

05/2021 – 08/2021 *Kampen, The Netherlands*

Investigated the quality trade-offs of cross-silo horizontal federated learning for industrial machinery manufacturers via a systematic literature review. I designed and validated a framework for navigating competing requirements (privacy, communication efficiency) through a UTAUT-based case study.

LIDL Netherlands Headquarters

Customer Service Representative

07/2018 – 03/2019 *Huizen, The Netherlands*

Handled 100+ requests per day by phone and email, resolving service issues at LIDL's national headquarters.

Education

University of Amsterdam, Master in Artificial Intelligence
09/2021 – 08/2024

- Research-centered program — 3.8/4.0 GPA
- Thesis: Synthesizing Object Dynamics from Human Motion with a Video Generative Prior (8/10)

University of Twente, Bachelor in Technology and Liberal Arts & Sciences
09/2018 – 08/2021

- Focused on machine learning and data science — 4.0/4.0 GPA
- Honours Programme: Processes of Change (additional 30 ECs)
- Thesis: Non-functional Requirements, Solutions and Interdependencies in Cross-Silo Horizontal Federated Learning

Publications

- **InterDyn: Controllable Interactive Dynamics with Video Diffusion Models** Paper | Website
  ***R. Akkerman**, H. Feng, M. J. Black, D. Tzionas, V. Fernández Abrevaya*
  in Computer Vision and Pattern Recognition (**CVPR**), 2025
- **OFER: Occluded Face Expression Reconstruction** Paper | Website
  *P. Selvaraju, V. Fernández Abrevaya, T. Bolkart, **R. Akkerman**, T. Ding, F. Amjadi, I. Zharkov*
  in Computer Vision and Pattern Recognition (**CVPR**), 2025
- **[Re] Explaining in Style: Training a GAN to explain a classifier in StyleSpace** Paper
  *N. Van der Vleuten, T. Radusinović, **R. Akkerman**, M. Reksoprodjo (equal contribution)*
  in ReScience C, 8(2), 2022; ML Reproducibility Challenge (**MLRC**) 2021

Volunteering

CVPR & Eurographics
Peer Reviewer
2025 – 2026
Reviewed conference submissions in computer graphics and computer vision.

University of Twente Innovation Fellows
Fellow / Coordinating Team
10/2019 – 03/2024 *Enschede, The Netherlands*
Directed a team of 12 students, overseeing strategic campus initiatives and co-managing an annual operating budget of €60,000; organised the 24-hour Twente-4-Hours event (KLM, Strategyzer, Stanford, FH Salzburg) and mentored new Fellows cohorts at TU Hamburg and Erasmus Rotterdam through Design Thinking training.

DIH-HERO
Grants & Contracts Support Officer
02/2021 – 08/2021 *Enschede, The Netherlands*
Supported coordination of a €16M Horizon 2020 action building a pan-European network of healthcare-robotics Digital Innovation Hubs across 17 partners, contributing to communications, events, and operations.

Autumn Challenge
Coach
08/2020 – 01/2021
Advised an international student team developing a Care Hotel concept report.

Stichting Huizer Botters
Secretary, De Bramzijgers
01/2017 – 08/2018 *Huizen, The Netherlands*
Co-established "Bramzijgers", the foundation's youth wing, to pass on traditional botter craftsmanship and engage young members in preserving Huizen's maritime heritage.

Skills

- **Technical**: Python, Bash, PyTorch, Hugging Face Transformers, PyTorch3D (differentiable rendering), PyRender/EGL (headless cluster rendering), NumPy, OpenCV, SMPL-X & SMPL-family body models, project-aria; distributed multi-GPU training on A100/H100 clusters via HTCondor
- **Project & People**: Project management, cross-functional team leadership, budget ownership, event organisation, technical & scientific communication
- **Languages**: English (proficient), Dutch (native), Spanish (learning), German (beginner)